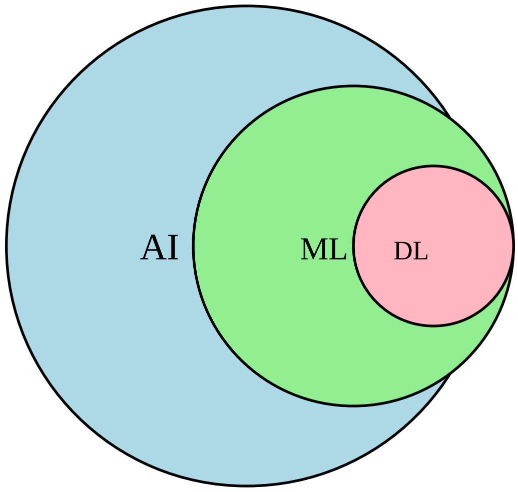
AI vs ML vs DL

Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) are closely related fields but differ in scope, complexity, and capabilities. Understanding the differences among them is crucial to grasp how modern intelligent systems are built and operate.



Artificial Intelligence (AI):
AI is a broad concept that refers to machines designed to perform tasks that typically require human intelligence. These include reasoning, problem-solving, perception, language understanding, and decision-making.

Examples: Chatbots, Recommendation Systems, Self-driving Cars, Game-playing Bots.

Machine Learning (ML):
ML is a subset of AI that focuses on algorithms enabling machines to learn from data without explicit programming. It uses statistical methods to make predictions or decisions.

Examples: Email Spam Filtering, Stock Price Prediction, Image Recognition.

Deep Learning (DL):
DL is a specialized subset of ML based on artificial neural networks that mimic the human brain. It excels at handling large amounts of unstructured data like images, audio, and text.

Examples: Face Recognition, Voice Assistants, Autonomous Vehicles.

Feature	AI	ML	DL
Definition	Broad field of creating intelligent systems	Subset of AI enabling machines to learn from data	Subset of ML using neural networks
Data Dependency	Can work with less data	Requires structured data	Requires large unstructured data
Complexity	Moderate	High	Very High
Human Intervention	High	Medium	Low
Examples	Robotics, Game AI	Fraud Detection, Recommendations	Image & Speech Recognition

Summary:
AI is the overarching field, ML is a subset that enables machines to learn from data, and DL is a further subset that leverages neural networks for high-level abstraction and automation.